S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Post Lab

Aim: To develop a C program for an Automated Backup and Recovery System that periodically backs up important files and restores them in case of data loss or system failure.

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Aim

To design and implement an Automated Backup and Recovery System using C programming that can create backup copies of important files and restore them in case of system failure or data loss.

Objectives

- To understand the concept of file system management in Operating System
- To implement file handling operations in C
- To simulate data backup and recovery mechanisms
- To ensure data safety and reliability
- To study how OS manages data protection and restoration

Requirements

Hardware:

- Computer/Laptop

Software:

- C Compiler (e.g., GCC Compiler)
- Operating System (Windows/Linux such as Ubuntu)

Theory

An Automated Backup and Recovery System is a mechanism used to protect data from loss due to system crashes, accidental deletion, or hardware/software failures. It is an important feature provided by operating systems to ensure data integrity and reliability.

Backup

Backup refers to the process of creating a duplicate copy of data and storing it in a separate location so that it can be used later if the original data is lost.

Recovery

Recovery is the process of restoring data from backup when the original data becomes unavailable due to failure.

Role of Operating System

The Operating System plays a key role in:

- Managing files and directories
- Providing file access permissions
- Handling storage devices

- Ensuring data consistency
- Supporting backup and recovery utilities

Working Principle

1. The system reads data from the original file.
2. It creates a backup file and copies all contents into it.
3. In case of failure, the system reads from the backup file.
4. It restores the data into a new or original file.

Advantages

- Prevents permanent data loss
- Ensures data availability
- Improves system reliability
- Useful in real-world applications like cloud storage and databases

Code:

```
#include <stdio.h>
#include <stdlib.h>

void backupFile() {
    FILE *source, *backup;
    char ch;

    source = fopen("original.txt", "r");

    if (source == NULL) {
        printf("Error: Original file not found!\n");
        return;
    }

    backup = fopen("backup.txt", "w");

    if (backup == NULL) {
        printf("Error: Cannot create backup file!\n");
        fclose(source);
        return;
    }

    while ((ch = fgetc(source)) != EOF) {
        fputc(ch, backup);
    }

    printf("Backup created successfully.\n");

    fclose(source);
```

```
    fclose(backup);
}

void recoverFile() {
    FILE *backup, *recovered;
    char ch;

    backup = fopen("backup.txt", "r");

    if (backup == NULL) {
        printf("Error: Backup file not found!\n");
        return;
    }

    recovered = fopen("recovered.txt", "w");

    if (recovered == NULL) {
        printf("Error: Cannot create recovery file!\n");
        fclose(backup);
        return;
    }

    while ((ch = fgetc(backup)) != EOF) {
        fputc(ch, recovered);
    }

    printf("File recovered successfully.\n");

    fclose(backup);
    fclose(recovered);
}

int main() {
    int choice;

    do {
        printf("\n--- Automated Backup & Recovery System ---\n");
        printf("1. Backup File\n");
        printf("2. Recover File\n");
        printf("3. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                backupFile();
                break;
            case 2:   recoverFile();
```

```
            break;
        case 3:
            printf("Exiting program...\n");
            break;
        default:
            printf("Invalid choice! Try again.\n");
    }

} while(choice != 3);

return 0;
}
```

Output:

Output

```
--- Automated Backup & Recovery System ---
1. Backup File
2. Recover File
3. Exit
Enter your choice: 1
ERROR!
Error: Original file not found!

--- Automated Backup & Recovery System ---
1. Backup File
2. Recover File
3. Exit
Enter your choice: 2
ERROR!
Error: Backup file not found!

--- Automated Backup & Recovery System ---
1. Backup File
2. Recover File
3. Exit
Enter your choice: 4
Invalid choice! Try again.

--- Automated Backup & Recovery System ---
1. Backup File
2. Recover File
3. Exit
Enter your choice: 3
Exiting program...


=== Code Execution Successful ===
```

Conclusion:
This experiment successfully demonstrates the concept of an Automated Backup and Recovery System using C programming. It highlights the importance of file handling, data protection, and recovery mechanisms in an Operating System. The system ensures that data can be safely backed up and restored, improving reliability and preventing data loss.

Discussion Questions

1. What is the importance of backup in an operating system?
2. How does recovery help in maintaining system reliability?
3. What are the limitations of this basic backup system?
4. How can this system be automated in real-time operating systems?
5. What are advanced backup techniques used in modern operating systems?

References:
Operating System Concepts – Silberschatz
The C Programming Language – Kernighan & Ritchie
GeeksforGeeks (C File Handling)